

Student ID:

Name:

* This assignment is prepared based on Chapter 9 (Chiller) and Chapter 11 (Heat Pump) among the class materials

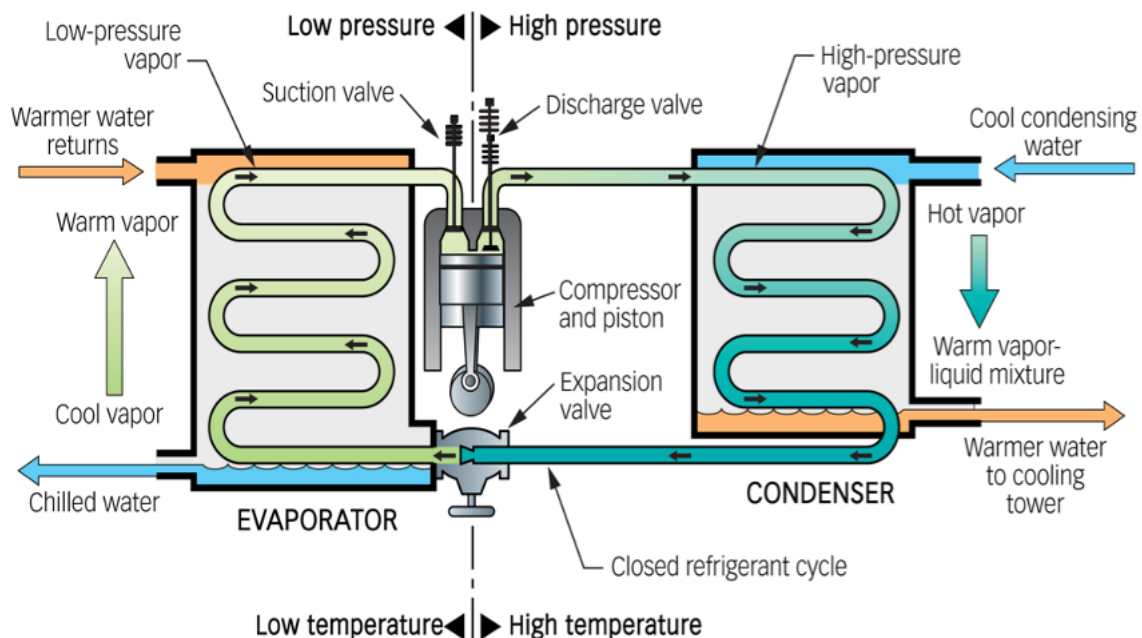
* While carrying out the assignment, if there are parts you learned individually besides the contents covered in class, the lecturer recommends you describe them freely and **plans to differentially grant bonus points (reflected in the total semester score) according to the degree of fidelity of additional study in terms of independent learning.**

* Please note that there are no restrictions on using generative AI (e.g., ChatGPT, Claude, Gemini) for your additional study (i.e., independent learning). However, if you do so, please include the prompts you used and **explain how you reviewed the AI-generated responses.**

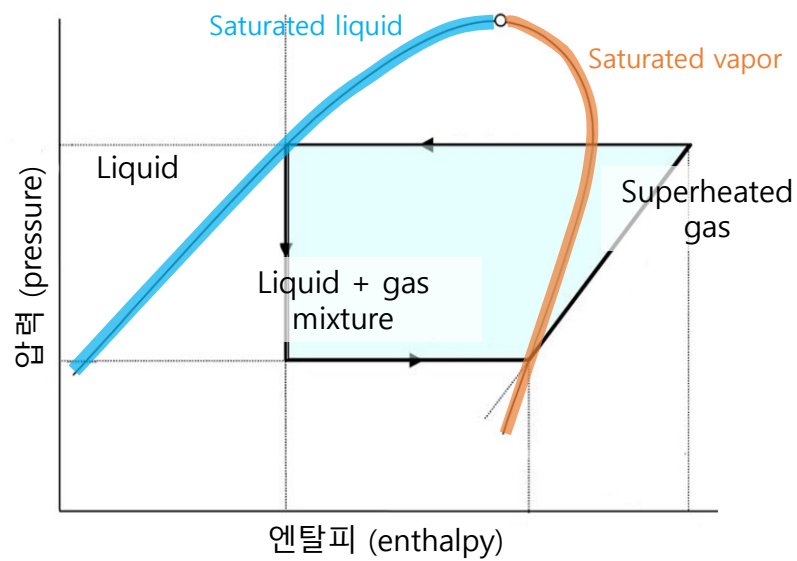
* You may freely choose the document format (MS Word, MS PowerPoint, HWP, or handwritten). Please convert it to a PDF format and submit it via eCampus.

* There are 2 questions in total, worth a combined 100 points.

1. [Principle of chiller] Figure 1-a is an example of the chiller configuration, and Figure 1-b shows the chiller refrigerant cycle on the "Mollier Diagram", a refrigerant state analysis tool (50 points).



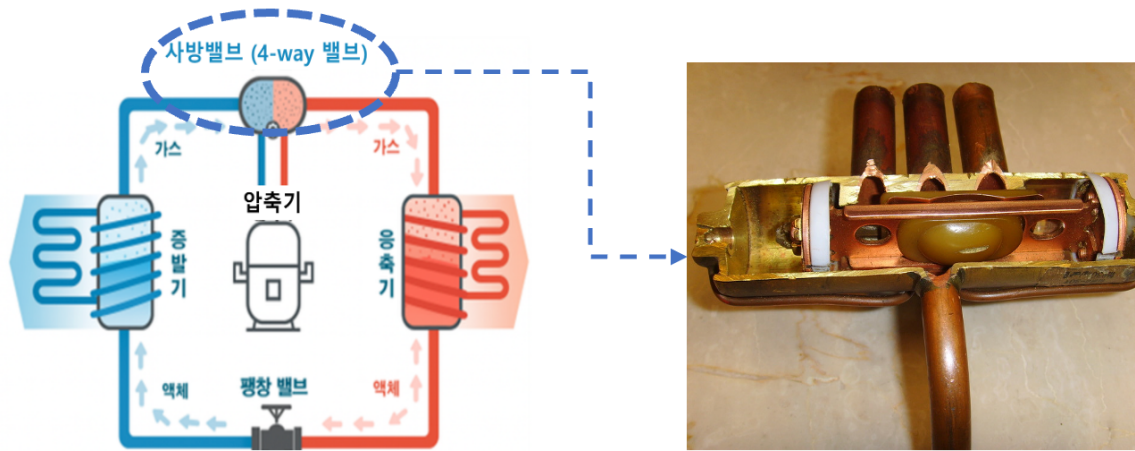
(Figure 1-a) Chiller configuration



(Figure 1-b) Mollier diagram and refrigerant cycle

- 1) Please describe the 4 processes of the refrigerant cycle in order (10 points)
 ([??] → [??] → [??] → [??])
- 2) Please mark the position corresponding to the 1st process on the Mollier diagram, and explain the chiller components and mechanisms involved in that process (10 points)
- 3) Please mark the position corresponding to the 2nd process on the Mollier diagram, and explain the chiller components and mechanisms involved in that process (10 points)
- 4) Please mark the position corresponding to the 3rd process on the Mollier diagram, and explain the chiller components and mechanisms involved in that process (10 points)
- 5) Please mark the position corresponding to the 4th process on the Mollier diagram, and explain the chiller components and mechanisms involved in that process (10 points)

2. [Heating and cooling principle of heat pump] Figure 2 shows the heat pump configuration (50 points).



(Figure 2) Heat pump configuration (left) and 4-way valve (right)

- 1) Please explain the refrigerant flow inside the 4-way valve during heating mode (20 points)
- 2) Please explain the refrigerant flow inside the 4-way valve during cooling mode (20 points)
- 3) Please explain why the heat pump is being considered as a renewable energy resource (10 points)